

City University of Hong Kong 2025-2026
Semester A CS3343 Software Engineering
Practice

Project Plan
Group 19
Classic UNO Card Game

Conducted By:

Woo King Hei 58533231

Wong Ka Chung 58533673

Chan Pak Yuen 58535918

Chan Wing Ki 58111083

Ho Sung Hin 58010837

Wong Hei Leong 58537010



1. Summary.....	3
1.1 Project Description.....	3
1.2 Stakeholders.....	3
1.3 Objectives of the Project.....	3
1.4 Project Requirements.....	4
1.5 Procurement (Outsourcing).....	4
2. Software Development Methodology.....	5
2.1 Project Team Organization.....	5
2.2 Development Tools.....	5
2.2.1 Development IDE Software.....	5
2.2.2 Development Platform.....	5
2.2.3 Test Cases.....	6
2.2.4 Coverage Checking.....	6
2.2.5 Testing Platform.....	6
2.2.6 Documentation.....	6
2.2.7 Version Control.....	6
3. Workload Breakdown Structure.....	7
4. Project Schedule.....	8

1. Summary

1.1 Project Description

UNO has been a well known card game around the globe for years. The simple game rules and fun, interactive gameplay are the main reasons why it has been so popular since its release. However, some players may have difficulty finding friends and suitable locations to play UNO.

Therefore, our group has developed a virtual version of UNO, which allows players to play UNO with CPU Players anywhere they want. This way, they can enjoy the game they love whenever they want, regardless of the settings.

1.2 Stakeholders

Stakeholder	Description
User	The users are the ones using our application to play UNO. They can also provide feedback for the developers to improve the game.
Developers	The developers are the team members responsible for maintaining the operation of the game system and launching future updates.

1.3 Objectives of the Project

The main objectives of the project are:

- **Virtual Version of UNO**

The main objective of this project is to develop a virtual version of UNO. We believe there is a demand for a UNO that players can play anywhere regardless of their location.

- **Convenient and User Friendly**

We also put emphasis on optimising user experience. We achieve this by providing a clean and easy-to-navigate user interface. Users can start a new game, play cards, and use the functions in our game without facing any confusion or difficulty.

- **Single Player Viability**

We understand that some people may have trouble finding other peers to play UNO with, therefore, our game is for single player, and players can play UNO with CPU players. Not only do they get to enjoy

UNO, they can also try to analyse the playing patterns of CPU players, hence adding another layer of challenge into the game.

1.4 Project Requirements

The functional requirement(s) in this project include:

1. **Start new game**

Once players open our game, they will be directed to the home page, where they can start a new game, playing with CPU players.

2. **All available actions in UNO**

During the game, users are able to perform actions in a UNO game such as:

Playing a Card

Drawing Card(s)

Shouting “UNO”

Catching Other Players Who Forgot to Call UNO

Choosing color, etc.

3. **CPU Players**

Players will be playing with CPU players with fixed playing logic.

The non-functional requirement(s) in this project include:

1. **User Friendly Interface**

To optimize playing experience, the game provides an organised user interface for users to play the game of UNO without any inconvenience or difficulties regarding navigating the app.

1.5 Procurement (Outsourcing)

In order to focus our core business which is the UNO product program designing, we decide to plan procurement for the card images design:

Reference: <https://github.com/john-costanzo/uno-card-images>

*Note: we added a colored version of wild card and wild draw four cards by editing the original cards with image-editing tools.

2. Software Development Methodology

When developing this UNO project, a divide-and-conquer approach was used. In the early stages of the project, we defined the requirements and corresponding functional components of the game, such as playing cards, shouting "UNO", etc. Our group then divided up the components for each team member to develop and test independently. After that, we integrated the code and improved code quality over time via refactoring and future updates.

2.1 Project Team Organization

Name	Title	Description
Wong Hei Leong	Coordinating Project Manager	PM, Development, Quality Control
Chan Pak Yuen	Developing Analyst	Test Cases, Player Demo
Ho Sung Hin	Software Architect	Development, Test Cases
Wong Ka Chung	Testing Engineer	Prototyping, Program Design
Chan Wing Ki	Analyst Programmer	Development, Code Refactoring
Woo King Hei	Project Manager	PM, Documentation, Bug Testing

2.2 Development Tools

2.2.1 Development IDE Software

- Eclipse IDE for Java Developers - 2024-09
- Visual Code Studio

2.2.2 Development Platform

- macOS Monterey 12.0.1 (x86_64)

2.2.3 Test Cases

- JUnit v5

2.2.4 Coverage Checking

- Eclipse IDE built-in coverage runner

2.2.5 Testing Platform

- macOS Monterey 12.0.1 (x86_64)

2.2.6 Documentation

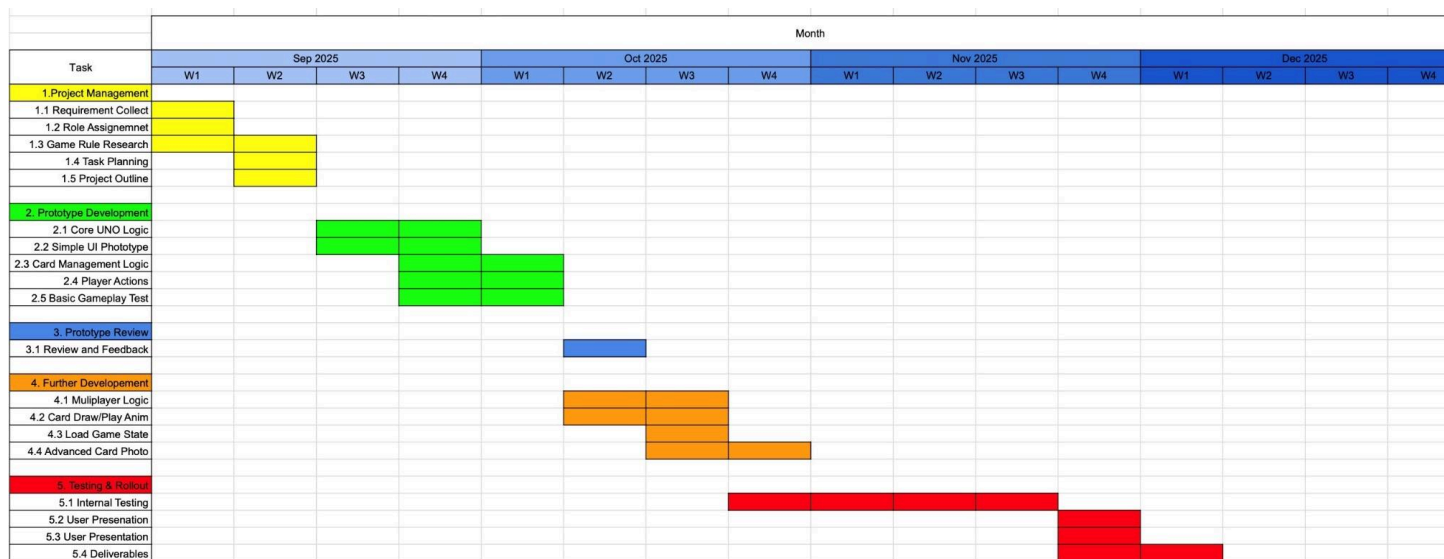
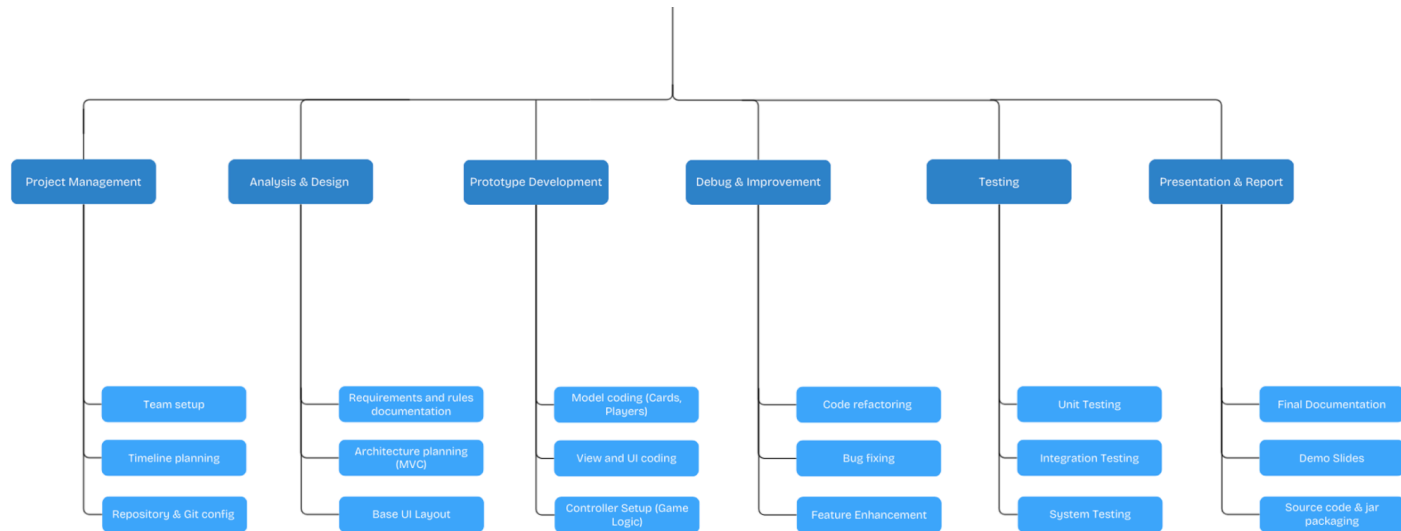
- Visual Paradigm 16.2 - Use Case Diagram, Class Diagram and Sequence Diagram
- Bugzilla - Bug Report
- Google Doc - Report

2.2.7 Version Control

Our group used GitHub for collaboration and version control. GitHub enables our team members to develop the code independently on their respective branches without affecting the main branch. Using GitHub also helped our team to track progress and changes due to updates. Overall, github greatly boosted our collaboration and efficiency when developing the code.

3. Workload Breakdown Structure

In terms of workload distribution, we divided different parts of development and assigned each of them to different teammates. Each development phrase can be broken down into details and smaller internal stages. Therefore, a tree structure is used to better organize the workload structure, as shown below.



4. Project Schedule

Task	Start Date	End Date
1. Project Management		
1.1 Requirement Collect	01/09/2025	03/09/2025
1.2 Role Assignment	01/09/2025	04/09/2025
1.3 Game Rule Research	03/09/2025	08/09/2025
1.4 Task Planning	08/09/2025	12/09/2025
1.5 Project Outline	10/09/2025	12/09/2025
2. Prototype Development		
2.1 Core UNO Logic	13/09/2025	24/09/2025
2.2 Simple UI Prototype	17/09/2025	27/09/2025
2.3 Card Management Logic	20/09/2025	01/10/2025
2.4 Player Actions	25/09/2025	03/10/2025
2.5 Basic Gameplay Test	28/09/2025	04/10/2025
3. Prototype Review		
3.1 Review and Feedback	04/10/2025	07/10/2025
4. Further Development		
4.1 Multiplayer Logic	08/10/2025	18/10/2025
4.2 Card Draw/Play Anim	12/10/2025	22/10/2025
4.3 Load Game State	18/10/2025	23/10/2025
4.4 Advanced Card Photos	21/10/2025	28/10/2025
5. Testing & Rollout		
5.1 Internal Testing	29/10/2025	19/11/2025
5.2 Polish & UI Update	20/11/2025	27/11/2025
5.3 User Presentation	28/11/2025	28/11/2025
5.4 Deliverables	28/11/2025	8/12/2025